

Views

```
-- Sales = Quantity * UnitPrice
-- CustomerID, sales
```

```
CREATE OR REPLACE VIEW `pareto-4.paretodemo.sales_v1`
AS
SELECT
    CustomerID,
    (Quantity * UnitPrice) AS sales
FROM `pareto-4.paretodemo.sales`;
```

```
-- GROUP BY customers, sales for each customer in one row of data
```

```
CREATE OR REPLACE VIEW `pareto-4.paretodemo.sales_v2`
AS
SELECT
    CustomerID,
    SUM(sales) AS customer_revenue
FROM `pareto-4.paretodemo.sales_v1`
GROUP BY CustomerID;
```

```
-- CustomerID, customer_revenue, cum_customers, total_customers, cum_revenue, total_revenue
-- View3
```

```
CREATE OR REPLACE VIEW `pareto-4.paretodemo.sales_v3`
AS
SELECT
    CustomerID,
    customer_revenue,
    ROW_NUMBER() OVER(ORDER BY customer_revenue DESC) AS cum_customers,
    COUNT(*) OVER() AS total_customers,
    SUM(customer_revenue) OVER(ORDER BY customer_revenue DESC
        ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS cum_revenue,
    SUM(customer_revenue) OVER() AS total_revenue
FROM `pareto-4.paretodemo.sales_v2`;
```

```
-- CustomerID, customer_revenue, cum_customers, total_customers, cum_revenue, total_revenue
-- + cum_sales_share, cum_pct_customers
-- View4
```

```
CREATE OR REPLACE VIEW `pareto-4.paretodemo.sales_v4`
AS
SELECT
```

```
CustomerID,  
customer_revenue,  
cum_customers,  
total_customers,  
cum_revenue,  
total_revenue,  
cum_revenue / total_revenue AS cum_sales_share,  
cum_customers / total_customers AS cum_pct_customers  
FROM `pareto-4.paretodemo.sales_v3`;
```

Views Final

```
-- declare variable
DECLARE target_sales_pct FLOAT64 DEFAULT 0.80;

-- number_of_customers, cum_revenue, total_revenue, target_sales_percent,
-- target_sales, cum_sales_share, cum_pct_customers
SELECT
  MIN(cum_customers) AS number_of_customers,
  MIN(total_customers) AS total_customers,
  MIN(cum_revenue) AS cum_revenue,
  MIN(total_revenue) AS total_revenue,
  target_sales_pct * 100 AS target_sales_percent,
  MIN(total_revenue * target_sales_pct) AS target_sales,
  MIN(cum_sales_share) AS cum_sales_share,
  MIN(cum_pct_customers) AS cum_pct_customers
FROM `pareto-4.paretodemo.sales_v4`
WHERE cum_sales_share >= target_sales_pct;
```

CTEs

```
-- DECLARE
```

```
DECLARE target_sales_pct FLOAT64 DEFAULT 0.60;
```

```
WITH base_sales AS (  
  SELECT  
    CustomerID,  
    (Quantity * UnitPrice) AS sales  
  FROM `pareto-4.paretodemo.sales`  
) ,  
customer_sales AS (  
  SELECT  
    CustomerID,  
    SUM(sales) AS customer_revenue  
  FROM base_sales  
  GROUP BY CustomerID  
) ,  
ranked AS (  
  SELECT  
    CustomerID,  
    customer_revenue,  
    ROW_NUMBER() OVER(ORDER BY customer_revenue DESC) AS cum_customers,  
    COUNT(*) OVER() AS total_customers,  
    SUM(customer_revenue) OVER(ORDER BY customer_revenue DESC  
      ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS cum_revenue,  
    SUM(customer_revenue) OVER() AS total_revenue  
  FROM customer_sales  
) ,  
with_pct AS (  
  SELECT  
    CustomerID,  
    customer_revenue,  
    cum_customers,  
    total_customers,  
    cum_revenue,  
    total_revenue,  
    cum_revenue / total_revenue AS cum_sales_share,  
    cum_customers / total_customers AS cum_pct_customers  
  FROM ranked  
)  
SELECT  
  MIN(cum_customers) AS number_of_customers,  
  MIN(total_customers) AS total_customers,
```

```
MIN(cum_revenue) AS cum_revenue,  
MIN(total_revenue) AS total_revenue,  
target_sales_pct * 100 AS target_sales_percent,  
MIN(total_revenue * target_sales_pct) AS target_sales,  
MIN(cum_sales_share) AS cum_sales_share,  
MIN(cum_pct_customers) AS cum_pct_customers  
FROM with_pct  
WHERE cum_sales_share >= target_sales_pct;
```